Muhammad Sohaib Ayub

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Professional Profile

Postdoctoral Researcher at Data Science Institute, University of Galway, with a research programme spanning web sustainability, accessibility, and data spaces. Current work focuses on energy-efficient web technologies, carbon-aware digital design, web accessibility (WCAG/universal design), and EU data infrastructure, with contributions to the European Commission-funded Data Spaces Support Centre (DSSC) the GreenWebInspector project accepted at ACM Web Conference 2026, a multi-partner Springer book chapter on designing data spaces, and a crowdsourced accessibility remediation framework accepted at the 8th International Universal Design Conference (UD26).

Track record includes 36+ peer-reviewed publications across Q1 journals (Scientific Data IF 6.9, Scientific Reports IF 3.9, IEEE Access IF 3.6) and A/A*-ranked venues (ICSE, ISSRE), 300+ citations, and active international collaborations across Ireland, New Zealand, China, Czech Republic, and Saudi Arabia. Active in European research networks as a member of two COST Actions (BOPALiM CA23101 and ARiF CA23135), chair and organiser of the SE4ALL workshop at ICSE 2027, and a member of the ICSE 2027 Shadow Programme Committee. Secured over EUR 40K in competitive funding as Co-PI across two LUMS Faculty Initiative Fund grants. Committed to open, reproducible research and FAIR data principles, with datasets and tools deposited publicly on GitHub and Zenodo.

Over 10 years of teaching experience across Ireland and Pakistan, spanning introductory programming to advanced algorithms, with more than 150 students advised in senior year projects. Currently completing the MAP4 Postgraduate Diploma in Academic Practice (CELT, University of Galway) and hold a Faculty Certificate in Teaching and Learning from LUMS, with demonstrated expertise in universal design for learning, research-led pedagogy, and inclusive assessment aligned with the Irish National Framework of Qualifications.

Research Vision & Interests

My research sits at the intersection of **web engineering**, **sustainability**, and **accessibility** – three areas that I argue must be addressed together if the web is to serve all users equitably without disproportionate environmental cost. I am building a programme around three interconnected themes:

- **Green Web & Carbon-Aware Digital Design.** I develop methods and tools for measuring, modelling, and reducing the energy footprint of web applications. GreenWebInspector provides real-time energy profiling for web pages; ongoing work extends this to carbon-aware design guidelines for civil society and under-resourced organisations. Future directions include automated energy regression testing integrated into CI/CD pipelines, and policy-facing metrics for the EU's digital sustainability agenda.
- **Web Accessibility & Universal Design.** My WARD project audited 103 rare disease patient organisation websites across five countries, finding systemic WCAG failures that exclude the most vulnerable users. I am developing automated and crowdsourced remediation frameworks for organisations without dedicated accessibility teams, the first of which was accepted at the 8th International Universal Design Conference (UD26) [3]. This work connects to Ireland's National Disability Authority mandate and the EU Web Accessibility Directive, creating clear pathways for Research Ireland and Horizon Europe funding.
- **Data Spaces & Interoperability.** Through the EU-funded DSSC, I contribute to the infrastructure enabling trustworthy, FAIR data sharing across European sectors. My interest is in the governance, energy, and accessibility dimensions of data space design – a relatively unexplored intersection with significant policy relevance for Ireland's role in European digital infrastructure.

I am ready to supervise PhD students in web sustainability, accessibility engineering, and applied ML for digital systems, building on an established co-supervision track record of 8 co-authored publications with doc-

toral researchers at KADE Lab, LUMS. I am well positioned to attract funding from **Research Ireland** (Frontiers for the Future, Starting Investigator Research Grant), **IRC** (Laureate), **Horizon Europe** (Cluster 4: Digital, Industry and Space), and **NGI** programmes, given my active EU project involvement, track record of international collaboration, and publication record in high-impact venues.

Education

PhD Computer Science 2023
Lahore University of Management Sciences (LUMS), Lahore, Pakistan

- Thesis: "Data-driven approaches for context-aware sports performance assessment and web energy optimization"
- Supervisor: Dr. Imdad Ullah Khan, Associate Professor, LUMS
- Key Courses: Big Data Analytics, Network Security, Internet of Things, Multicore Computing, Human-Computer Interaction Design
- Fully Funded PhD Fellowship – PKR∼ 7M (∼EUR 23K)
- LUMS Travel Grant to present at ICSE (2018), Gothenburg, Sweden – PKR ∼1M (∼EUR 3.3K)

MS Computer Science 2015
Lahore University of Management Sciences (LUMS), Lahore, Pakistan

- Thesis: "Efficiently Finding Minimal Failing Input in Map-Reduce Programs"
- Supervisor: Dr. Junaid Haroon Siddiqui, Associate Professor, LUMS
- Key Courses: Machine Learning, Reliable Software, Distributed Systems, Topics in Internet Research, Service Oriented Computing, Computer Architecture

BS Information Technology 2012
National University of Science and Technology (NUST), Islamabad, Pakistan

- Final Year Project: "Design and Development of a Java Parallel I/O Library" [38]
- Key Courses: Web Technologies, Computing Algorithms, Operating Systems, Database Systems, Software Engineering, Data Warehousing & Data Mining, Enterprise Network Technologies (Linux)
- Fully funded Prime Minister's National ICT-R&D Scholarship – PKR ∼1.2M (∼EUR 4K)

Research Experience

Postdoctoral Researcher May 2025–Present
Data Science Institute, University of Galway, Ireland

- **Data Spaces.** Worked on the EU-funded Data Spaces Support Centre (DSSC), where the challenge is enabling organisations across Europe to share data in a trustworthy and interoperable way. Co-authored project deliverables and published research, and helped shape design principles and best practices now guiding how data spaces are built across sectors. This work is consolidated in a multi-partner book chapter on principles and practices for designing data spaces [5], accepted for the Springer volume *Data Spaces: Enabling Trusted Data Ecosystems (Season 2)*, alongside a journal paper on data space best practices currently in preparation.
- **Web Energy.** The web is a growing and largely invisible source of carbon emissions. Built a predictive approach to estimate the energy a website consumes and turned it into a real-time assessment tool, accepted at ACM Web Conference 2026 (Core A*) [4], giving developers a practical way to measure and cut the footprint of what they build.
- **Health & Accessibility.** Established a research line on digital equity for rare disease patients, who face some of the steepest barriers to reliable online health information. Assessed the quality and AI readiness of patient health communities, exposed systemic accessibility failures through a five-country website audit where not a single site met compliance standards, and proposed a crowdsourcing approach so under-resourced patient organisations can fix these barriers at scale, accepted at the 8th International Universal Design Conference (UD26) [3].

- **European Networks & Community Building.** Active member of two COST Actions, BOPALiM (CA23101) on accessibility and community mobility and ARiF (CA23135) on augmented reality in forestry, contributing across five working groups; ARiF connects directly to my Pakistan Forest Observatory work [1], a satellite, IoT and ML platform built over two LUMS Faculty Initiative Fund cycles. Chairing and organising SE4ALL, the International Workshop on Software Engineering and AI for All, at ICSE 2027 in Dublin, and serving on the ICSE 2027 Shadow Programme Committee.
- **Funding Proposals.** Actively engaged in competitive proposal writing, supporting my advisor in coordinating Horizon Europe proposals including CRADLE (HORIZON-CL2-2026-01-DEMOCRACY-10), on AI-mediated disinformation and citizen behaviour, for which I drafted the state-of-the-art and literature review, and HORIZON-CL2-2026-01-HERITAGE-03, and preparing and submitting applications to the Research Ireland Government of Ireland Postdoctoral Fellowship (GOIPD), Enterprise Ireland Sustain-FIT (MSCA COFUND), NGI Zero Commons Fund (EU cascade funding), Research Ireland Public Service Fellowship, and the SEAI Research, Development and Demonstration (RD&D) Funding Programme.

PhD Research Associate Jan 2021–Sep 2023

Data Analytics Lab, LUMS, Lahore, Pakistan

- Predictive modelling of website power consumption [14] (under review at IEEE Transactions on Sustainable Computing) and cricket player performance analytics – CAMP metric [22], Journal of the Operational Research Society, Q1
- 3D Wireless Sensor Networks Localisation [25] – collaboration with Sun Yat-sen University, China; published IEEE Access, Q1
- Arabic OCR for handwritten manuscripts [24], [31] – collaboration with Islamic University of Madinah
- Synthetic browsing histories dataset for 50 countries [19] – collaboration with Brno University of Technology, Czech Republic; published Scientific Data, Q1
- Biological Data Analysis [32] in collaboration with Department of Life Sciences @ LUMS.
- Applied ML projects in taxi demand prediction [10], GenAI for contact centres [13], facial emotion detection [21], legal case outcome prediction [28], ordinal data imputation [29], and global optimization algorithms [30] – collaboration with Massey University, New Zealand
- Mentoring PhD students (KADE Lab, LUMS) on Urdu NLP: text summarisation [16], [17], [23], POS tagging [18], sentiment analysis [20], and cybercrime text analysis [26], [27]; resulting in 8 co-authored publications including Q1 journal papers

PhD Research Fellow Sep 2014–Dec 2020

Program Analysis Lab, LUMS, Lahore, Pakistan

- Software reliability and verification of MPI Java Programs using Software Model Checking [34], [35] – published at ISSRE (Core A, IEEE) and PPoPP (ACM)
- Delta debugging for MapReduce: efficiently finding minimal failing inputs [33] – published at ICSE-Companion (Core A*, IEEE)

Research Assistant Jul 2010–Sep 2012

High-Performance Computing Lab, SEECS NUST, Islamabad,Pakistan

- Parallel I/O Library for MPJ Express(Java HPC) [37]– published IEEE PDCAT 2012
- Counterfeit medicine identification using QR Codes and mobile phones [39]– published IEEE HONET 2011
- Summer Research Internee (2010 & 2011): contributed to core MPJ Express development tasks and copy-edited a parallel computing textbook in LaTeX; profiled MPJ Express to identify and resolve performance bottlenecks

Professional Experience

Software Development Engineer Jan 2013-Aug 2013

xFlow Research Inc., Islamabad

- Developed software-defined networking applications, enhancing network management and flexibility. [36]

Software Engineer Jul 2012-Jan 2013
Techlogix, Islamabad

– Delivered software support for Oracle FLEXCUBE, ensuring seamless operations across all Askari Bank branches.

Teaching Experience

Over 10 years of teaching across the University of Galway (Ireland), LUMS, and FAST-NU (Pakistan), covering courses from introductory programming to advanced algorithms. Co-advised more than 30 senior year projects involving over 150 students. Have designed and delivered full courses as sole instructor at LUMS and FAST-NU, and collaborated on course design at the University of Galway.
Teaching Philosophy: I connect theory to real-world data and societal problems, using scaffolded problem-solving, formative feedback, and universal design principles to support diverse cohorts. I am committed to equity in assessment and have engaged with inclusive teaching frameworks through the MAP4 PGDip (University of Galway) and the Faculty Certificate in Teaching & Learning (LUMS). I actively integrate research into undergraduate and postgraduate teaching.

Institution	**Course**	**Semester**	**Role**
UoG, Ireland	MS3106: Networks & Communication	Spring 2026	Guest Lecturer
LUMS, Lahore	CS 491A/B: Senior Year Projects Advising	2023–2025	Co-instructor
LUMS, Lahore	CS 202: Data Structures	Fall 2024	Co-instructor
LUMS, Lahore	CS 100: Computational Problem Solving	Spring 2024	Lead Instructor
LUMS, Lahore	CS 300: Advanced Programming	Spring 2024	Lead Instructor
LUMS, Lahore	CS 100: Computational Problem Solving	Fall 2023	Co-Instructor
LUMS, Lahore	CS 200: Introduction to Programming	Summer 2017	Lead Lab Instructor
LUMS, Lahore	CS 510: Design and Analysis of Algorithms	Spring 2015	Teaching Assistant
LUMS, Lahore	CS 582: Distributed Systems	Fall 2014	Teaching Assistant
LUMS, Lahore	CS 310: Algorithms	Spring 2014	Teaching Assistant
FAST-NU, Lahore	CS 205: Operating Systems	Spring 2015	Lead Instructor
NUST, Islamabad	CS 321: Enterprise Network Technologies/Linux	Spring 2012	Teaching Assistant
NUST, Islamabad	CS 251: Design and Development of Algorithms	Spring 2012	Teaching Assistant
NUST, Islamabad	ISE 312: Computing Algorithms	Spring 2011	Teaching Assistant
NUST, Islamabad	ISE 205: Object-Oriented Programming	Summer 2011	Teaching Assistant
NUST, Islamabad	ISE 103: Computer Fundamentals	Fall 2010	Teaching Assistant
NUST, Islamabad	EE 221/241: Digital Logical Design (6 semesters)	2010–2012	Teaching Assistant

Teaching & Learning Certifications and Advising

- MAP4 Postgraduate Diploma in Academic Practice | CELT, University of Galway 2025–ongoing
  – Key Modules: Research Supervision & Development, Teaching Online
- Universal Design Beyond the Classroom | AHEAD & UCD 2025
- Academic Advisor, BS Students, College of Science and Engineering | University of Galway 2025–2027
- Faculty Certificate in Teaching and Learning: Theory and Practice | LUMS Learning Institute 2024
  – Key Workshops: Instructional Skills Development, ChatGPT as Co-Instructor, Design Thinking for Online & In-Person Learning, Course Craft: Designing Effective Courses, From Crowd to Community: Best Practices in a Large Classroom, Authentic & Renewable Assessments, Flipping and Blending Classrooms, Inclusive Teaching Dialogue: Beyond Relative Grading

Funding, Grants & Awards

Received over EUR 43K (approx. PKR 13M) in competitive research grants, travel awards, and merit-based fellowships and scholarships.

Title	Year	Awarding Body	PKR	EUR
Travel Grant – WorldS4 2026, London	2026	University of Galway	–	∼EUR 3K
Co-PI, FIF Grant – *Forest Observatory*	2025–2026	LUMS	PKR 1.5M	∼EUR 5K
Co-PI, FIF Grant – *Forest Observatory*	2024–2025	LUMS	PKR 1.5M	∼EUR 5K
Travel Grant – ICSE 2018, Sweden	2018	LUMS	PKR ∼1M	∼EUR 3.3K
Travel Grant – PPoPP 2016, Spain	2016	LUMS	PKR ∼1.1M	∼EUR 3.4K
Fully Funded PhD Fellowship	2015–2019	LUMS	PKR ∼7M	∼EUR 23K
Fully Funded Bachelor's Scholarship	2008–2012	PM ICT&RD Fund	PKR ∼1.2M	∼EUR 4K

Community Service & Leadership

- **Workshop Chair & Organiser**, SE4ALL – International Workshop on Software Engineering and AI for All, co-located with ICSE 2027, Dublin, Ireland 2027
- **European Research Networks (COST Actions)**
  - CA23101 – BOPALiM, Building Opportunities for Participation and Accessibility through lifelong community Mobility (WG1, WG3) 2025–ongoing
  - CA23135 – ARiF, Bringing Digital Data and Reality Together: Augmented Reality in Forestry (WG1, WG3, WG4, WG5) 2025–ongoing
- **Service to Research Community** (50+ verified peer reviews – Web of Science)
  - **ACM Certified Reviewer** (ACM Reviewer Training and Certification) 2026
  - **Shadow Programme Committee:** ICSE 2027 2026–2027
  - **Journal Peer Reviews:** IEEE Transactions on Reliability, Journal of Big Data, IEEE Access, IJCDS, Jordanian Journal of Computers and IT
  - **Conference Programme Committee:** CHI, ACL Rolling Review (ARR), INTRACT, IEEE-3ICT, ADICS-ESIT, AICTC, AECT, IEEE-GCCCE, IEEE-ICOMITEE, EECSI, SCS-NCC, IEEE-ICCAIS, ICCAT, ISSPIT, IC-CISN, ICSNC, ICARCAD, ISCPDM, ICECET, ICST (Co-Reviewer), ISSRE (Co-Reviewer)
- **8 co-authored publications with supervised PhD students** at KADE Lab, LUMS, across Urdu NLP (text summarisation, POS tagging, sentiment analysis, cybercrime text analysis)
- **Academic Advisor**, BS Students, College of Science and Engineering, University of Galway 2025–2027
- **Residence Fellow, LUMS, Lahore, Pakistan** 2014–2017
  - Managed student housing and support services, fostering a supportive residential community
- **Volunteer for Outreach & DEI**
  - Public engagement talks on web accessibility and sustainability, University of Galway 2025–ongoing
  - Open Day, LUMS, Lahore 2015
  - Graduate Orientation, LUMS, Lahore 2014
  - Funfair for Special People, F9 Park, Islamabad 2012

Talks & Workshops

- Workshop Chair and Organiser, SE4ALL – International Workshop on Software Engineering and AI for All, ICSE 2027, Dublin, Ireland 2027

- Paper accepted for presentation at the 8th International Universal Design Conference (UD26), Galway, Ireland Nov 2026
- Presented the Agriculture Portal paper [2] at the WorldS4 Conference, London, UK 2026
- GreenWebInspector demo and poster at ACM Web Conference (WWW'26), Dubai, UAE 2026
- Presented a poster on delta debugging of MapReduce programs at ICSE, Gothenburg, Sweden 2018
- Student Volunteer, UX Pakistan Conference at LUMS 2017
- Facilitated a five-day Cloud Computing Workshop at LUMS as Teaching Assistant 2015
- Delivered a talk on High-Performance Computing at Askari Bank, Islamabad 2013
- Instructed a two-day web development workshop, HL7 Student Chapter, SEECS NUST 2011
- Supported organisation of a two-day Cloud Computing Workshop at SEECS NUST 2011

Skills

- **Programming:** Python, Java, C/C++, JavaScript, Haskell
- **Web & Accessibility Tools:** Google Lighthouse, axe-core, WAVE, Chrome DevTools, Chrome Extension development
- **Python Libraries:** pandas, numpy, scikit-learn, Hugging Face Transformers, BeautifulSoup, Selenium
- **Data & Infrastructure:** Data processing pipelines, Big Data Analytics, SQL, Git, LaTeX
- **Open Science:** Public repositories on GitHub, datasets on Zenodo, ORCID profile, preprints on ResearchGate; committed to FAIR data principles
- **Research Areas:** Web accessibility (WCAG, universal design), energy-efficient web technologies, carbon-aware digital design, data spaces, machine learning, NLP, sports analytics

Publications

36+ peer-reviewed publications including Q1 journal papers, a Springer book chapter, and A/A* conference papers (WWW, ICSE, ISSRE) with 300+ citations and collaborators across 6 countries. [Google Scholar]

[1] **M. S. Ayub**, S. Bakhat, M. M. Awais, A. M. Akhter, and U. u. Hassan, "Pakistan Forest Observatory: A Knowledge-Based Platform for AI-Driven Forest Monitoring, Risk Assessment, and Operational Management", 2026, Under review at AI-2026, the 46th SGAI International Conference on Artificial Intelligence, Cambridge, UK.

[2] **M. S. Ayub**, S. Bakhat, M. M. Awais, and U. u. Hassan, "Agriculture Portal: Sentinel-2 Spectral Indices and Ensemble Fuzzy Multi-Criteria Decision-Making for Crop Health, Soil Suitability, and Advisory Compliance", in *Intelligent Sustainable Systems: Selected Papers of WorldS4 2026, Lecture Notes in Networks and Systems*, London, UK: Springer, 2026.

[3] **M. S. Ayub** and U. u. Hassan, "A Crowdsourcing Framework for Web Accessibility Remediation in Under-Resourced Civil Society Organizations", in *Proceedings of the 8th International Universal Design Conference (UD26), Galway, Ireland*, IOS Press, 2026.

[4] **M. S. Ayub**, U. u. Hassan, I. U. Khan, and E. Curry, "GreenWebInspector: A Tool for Real-Time Web Energy Efficiency Assessment", in *Companion Proceedings of the ACM Web Conference*, **Core Rank: A***, 2026.

[5] **M. S. Ayub**, M. A. Razzaq, R. Haque, D. Alonso, M. Martin, C. Stolwijk, J. Suksi, M. Frigeri, M. Musidlowska, S. Rogotis, T. Tuikka, B. Homuth, S. De Panfilis, C. M. Pezuela-Robles, M. Al-Qatf, E. Curry, F. Ahmadi Zeleti, M. Rao, H. Korhonen, and S. Castellvi, "Principles and Practices for Designing Data Spaces", in *Data Spaces: Enabling Trusted Data Ecosystems, Season 2*, Springer, 2026.

[6] A. Faheem, F. Ullah, M. Hammad, A. Hassan, **M. S. Ayub**, and A. Karim, "U-MIRAGE: Benchmarking Chain-of-Thought Reasoning for Urdu Medical QA", in *The 2nd Workshop on NLP for Languages Using Arabic Script (AbjadNLP-EACL)*, **Core Rank: A**, 2026.

[7] U. u. Hassan, P. Paul, **M. S. Ayub**, K. Shaheed, and L. Hoang, "Data Sharing for Citizen-Driven Repair and Reuse of Electronics and Electrical Equipment", *Data in Brief*, 2026.

[8] D. Komosny, **M. S. Ayub**, and S. Rehman, "A Supervised System for Curating Browsing Whitelists for Individuals with Cognitive Disabilities under Legal Guardianship", *EPJ Data Science*, 2026, **JCR: Q1**.

[9] F. Ullah, F. Sikandar, A. Waqar, F. Ali, **M. S. Ayub**, M. Mushtaq, and A. Karim, "MaskedVerbalizer: Automatic Verbalizer Construction for Few-Shot Text Classification in Low-Resource Right-to-Left Languages", in *Language Resources and Evaluation (LREC)*, 2026.

[10] S. Alam, **M. S. Ayub**, H. Cui, and M. A. Khan, "A Comparative Study of Machine Learning Models for Taxi-Demand Prediction Using a Big Data Framework", *Public Transport*, pp. 1–31, 2025, **JCR: Q1**.

[11] S. Alam, I. K. Niazi, H. Shafi, W. A. Awan, I. Amjad, **M. S. Ayub**, and M. Mahmud, "Clustering-Based Balance Phenotyping in Older Adults from Treadmill Training Interventions", in *IEEE BigData*, 2025.

[12] S. Alam, A. Pawade, **M. S. Ayub**, S. U. Rehman, and **A. Ayub**, "Evaluating Explainable AI Implementation and User Agency Across Major Social Media Platforms", in *IEEE BigData*, 2025.

[13] S. Alam, L. Skelton, **M. S. Ayub**, and W. Dharmawan, "Generative Artificial Intelligence and Digital Transformation in Contact Center Businesses", in *International Conference on Information Resources Management (Conf–IRM)*, 2025.

[14] **M. S. Ayub**, N. Mustafa, S. Ali, J. Ahmad, I. Khan, M. A. Khan, and S. Faizullah, "Modeling Power Consumption of Web Apps using Static Design Features", 2025, Under review at IEEE Transactions on Sustainable Computing.

[15] A. Faheem, U. Faizad, **M. S. Ayub**, and A. Karim, "Roman Urdu Part of Speech (POS) Tagging", *Natural Language Engineering*, 2025, **JCR: Q1**.

[16] A. Faheem, U. Faizad, **M. S. Ayub**, and A. Karim, "Urdu Extractive Text Summarization: Evaluating Sentence Scoring Techniques", in *Applied Intelligence and Informatics (AII) Conference*, 2025.

[17] A. Faheem, F. Ullah, **M. S. Ayub**, and D. A. Karim, "Abstractive Summarization for Urdu Video Description Generation", *ACM Transactions on Asian and Low-Resource Language Information Processing*, 2025.

[18] A. Faheem, F. Ullah, U. Azam, **M. S. Ayub**, and A. Karim, "Part of speech (POS) tagging in Roman Urdu: datasets and models", *Language Resources and Evaluation*, pp. 1–28, 2025, **JCR: Q1**.

[19] D. Komosny, S. Rehman, and **M. S. Ayub**, "Synthetic Browsing Histories for 50 Countries Worldwide: Datasets for Research, Development, and Education", *Scientific Data*, 2025, **JCR: Q1**.

[20] F. Ullah, S. Faizullah, I. U. Khan, T. S. Ali, T. Alghamdi, **M. S. Ayub**, and A. Karim, "Prompt-Based Fine-Tuning with Multilingual Transformers for Language-Independent Sentiment Analysis", *Scientific Reports*, 2025, **JCR: Q1**.

[21] S. Alam, **M. S. Ayub**, R. Sathasivam, and M. A. Khan, "Enhanced Facial Emotion Detection Models Utilizing Geometry-Based Features for Superior Human-Computer Interaction", in *International Conference on Neural Information Processing (ICONIP)*, 2024.

[22] **M. S. Ayub**, N. Ullah, S. Ali, I. Khan, M. M. Awais, M. A. Khan, and S. Faizullah, "CAMP: A Player Performance Metric for Data-Driven Man of the Match Prediction in Cricket", *Journal of the Operational Research Society*, 2024, **JCR: Q1**.

[23] A. Faheem, F. Ullah, **M. S. Ayub**, and A. Karim, "UrduMASD: a Multimodal Abstractive Summarization Dataset for Urdu", in *Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*, ELRA, 2024.

[24] S. Faizullah, **M. S. Ayub**, T. Alghamdi, T. S. Ali, M. A. Khan, and E. Nabil, "Revolutionizing Historical Document Digitization: LSTM-Enhanced OCR for Arabic Handwritten Manuscripts", *International Journal of Advanced Computer Science and Applications (IJACSA)*, vol. 15, no. 10, 2024.

[25] M. Islam, M. Ikram, M. Alhussein, **M. S. Ayub**, M. A. Khan, and K. Aurangzeb, "A Component-Based Localization Algorithm for Sparse 3-D Wireless Sensor Networks", *IEEE ACCESS*, 2024.

[26] F. Ullah, **M. S. Ayub**, A. Faheem, M. M. Awais, and A. Karim, "Towards Unveiling the Potential of Fuzzy Values as Features: A Comparative Study in Cybercrime Text Analysis", in *International Conference on Neural Information Processing (ICONIP)*, 2024.

[27] F. Ullah, U. Azam, A. Faheem, **M. S. Ayub**, F. Kamiran, and A. Karim, "Detecting Cybercrimes in Accordance with Pakistani Law: Dataset and Evaluation using PLMs", in *Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*, ELRA, 2024.

[28] S. Alam, R. Pande, **M. S. Ayub**, and M. A. Khan, "Towards Developing an Automated Chatbot for Predicting Legal Case Outcomes: A Deep Learning Approach", in *Asian Conference on Intelligent Information and Database Systems (ACIIDS)*, Springer, 2023.

[29] S. Alam, **M. S. Ayub**, S. Arora, and M. A. Khan, "An Investigation of The Imputation Techniques for Missing Values in Ordinal Data Enhancing Clustering and Classification Analysis Validity", *Decision Analytics Journal*, vol. 9, p. 100 341, 2023, **JCR: Q1**.

[30] S. Alam, X. Zhao, I. K. Niazi, **M. S. Ayub**, and M. A. Khan, "A Comparative Analysis of Global Optimization Algorithms for Surface Electromyographic Signal Onset Detection", *Decision Analytics Journal*, vol. 8, p. 100 294, 2023, **JCR: Q1**.

[31] S. Faizullah, **M. S. Ayub**, S. Hussain, and M. A. Khan, "A Survey of OCR in Arabic Language: Applications, Techniques, and Challenges", *Applied Sciences*, vol. 13, no. 7, p. 4584, 2023.

[32] U. Sardar, S. Ali, **M. S. Ayub**, M. Shoaib, K. Bashir, I. U. Khan, and M. Patterson, "Sequence-Based Nanobody-Antigen Binding Prediction", in *International Symposium on Bioinformatics Research and Applications (ISBRA)*, Springer, 2023, pp. 227–240.

[33] **M. S. Ayub** and J. H. Siddiqui, "Efficiently Finding Minimal Failing Input in MapReduce Programs", in *International Conference on Software Engineering: Companion (ICSE-Companion)*, **Core Rank: A***, IEEE, 2018, pp. 177–178.

[34] **M. S. Ayub**, W. U. Rehman, and J. H. Siddiqui, "Experience Report: Verifying MPI Java Programs Using Software Model Checking", in *International Symposium on Software Reliability Engineering (ISSRE)*, **Core Rank: A**, IEEE, 2017, pp. 294–304.

[35] W. U. Rehman, **M. S. Ayub**, and J. H. Siddiqui, "Verification of MPI Java Programs using Software Model Checking", in *ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP)*, ACM, 2016, pp. 1–2.

[36] Q. Maqbool, **M. S. Ayub**, J. Zulfiqar, and A. Shafi, "Virtual TCAM for data Center Switches", in *Conference on Network Function Virtualization and Software Defined Network (NFV-SDN)*, IEEE, 2015, pp. 61–66.

[37] A. A. Awan, **M. S. Ayub**, A. Shafi, and S. Lee, "Towards Efficient Support for Parallel I/O in Java HPC", in *International Conference on Parallel and Distributed Computing, Applications and Technologies (PDCAT)*, IEEE, 2012, pp. 137–143.

[38] **M. S. Ayub**, M. Adnan, and M. Y. Shafi, "Design and Development of a Java Parallel I/O Library", Bachelor's Thesis, Department of Computing, SEECS, NUST, Islamabad, Pakistan, 2012.

[39] S. U. Rehman, R. U. Rasool, **M. S. Ayub**, S. Ullah, A. Kamal, Q. M. Rajpoot, and Z. Anwar, "Reliable Identification of Counterfeit Medicine Using Camera Equipped Mobile Phones", in *High Capacity Optical Networks and Enabling Technologies (HONET)*, IEEE, 2011, pp. 273–279.